4.8 Analyzing Graphs of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.8 · Student sheet

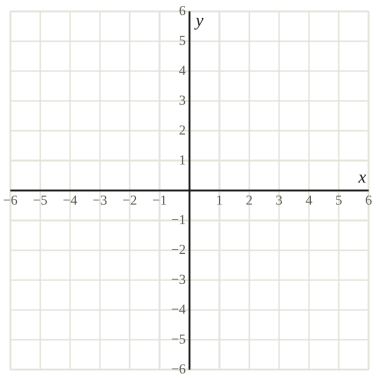
I can…

- use x-intercepts and end behavior to graph polynomial functions.
- find turning points and local maximums and minimums.
- tell whether a function is even, odd, or neither.

Part 1 · Sketch from factors

$y = (x + 2)(x - 1)(x - 3)$. x-intercepts: _____ End behavior: _____

Sketch first (no technology):



Now check in Desmos. Local maximum ≈ _____ Local minimum ≈ _____

Part 2 · Notes: fill in the blanks

A degree-n polynomial has at most _____ turning points.

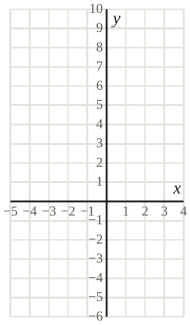
Even multiplicity: the graph _____ the x-axis. Odd: it _____.

Even function: symmetric about the _____. Odd function: symmetric about the _____.

Part 3 · Examples (work them with the class)

Example 1

Graph $f(x) = \frac{1}{2}(x + 3)(x - 2)^2$.



Example 2

Find all real zeros of $f(x) = x^3 - x^2 - 6x$.

Example 3

A polynomial graph has 3 turning points. What is its least possible degree?

Example 4

Tell whether each function is even, odd, or neither.

a. $f(x) = x^4 - 3x^2$ b. $g(x) = 2x^3 + x$ c. $h(x) = x^3 + 2$

Part 4 · Partner practice

Partner 1

Find all real zeros of $f(x) = x^4 - 9x^2$.

Partner 2

Is $f(x) = x^5 - 4x^3$ even, odd, or neither?